

Quick Check

Lesson 1: Anchoring Phenomenon Launch — "The Matatu Horn and the Fading Radio"

9 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Properties of Waves | Lesson 1

A matatu travelling in the opposite direction passes your bus on the Nakuru–Nairobi highway, and the driver does not change the horn at all. Which statement is an OBSERVATION (something you actually notice) rather than a guess about the cause?

- A** The horn changes its note because the matatu is tired after a long journey.
- B** The driver pressed the horn harder as the matatu came closer.
- C** The horn sounded higher-pitched as the matatu came towards you and lower-pitched as it moved away.
- D** The horn button on the matatu was faulty.

On the **PREDICT** card, the prompt 'What did you NOTICE?' asks for a fact about the bus journey. Which entry is a **NOTICE** and not a guess about the cause?

- A** The radio receiver was faulty.
- B** Classic 105 has a weaker transmitter than the AM station.
- C** On entering the Kabete underpass, the FM signal (Classic 105, 105.2 MHz) cut out almost completely while the AM signal stayed audible.
- D** AM waves are stronger than FM waves.

At Station A you use the PhET Sound simulation, first moving a listener past a stationary source and then moving the source past a stationary listener. What should you write in your observation table for each set-up?

- A** The speed of a matatu on the highway
- B** The perceived pitch (higher, lower or same) for that set-up
- C** The reason why the pitch changes
- D** The name of the scientific effect that causes the change

The Station B card shows the Kabete tunnel drawn to scale. The tunnel is about 8 m wide, the FM (105.2 MHz) wavelength is about 2.9 m and the AM wavelength is about 300 m. Which statement correctly reads these numbers?

- A** The AM wavelength is about 300 m, which is much longer than the tunnel width of about 8 m, while the FM wavelength of about 2.9 m is shorter than the tunnel width.
- B** Both the FM and the AM wavelengths are shorter than the tunnel width.
- C** The AM and FM wavelengths are about the same size as each other.
- D** The FM wavelength is longer than the tunnel width, and the AM wavelength is shorter than the tunnel width.

During the 'WHAT WE SAW' debrief, which sticky-note entry is a specific, measurable OBSERVATION that could go in the EVIDENCE column?

- A** The matatu horn is broken.
- B** FM wavelength is about 3 m and the tunnel is about 8 m wide.
- C** Classic 105 is a bad radio station.
- D** FM is blocked because the tunnel is too small.

During the gallery walk, each student places one dot sticker on the group explanation they find most convincing. Which explanation should get your dot?

- A** The one that blames a technology fault, such as 'Classic 105 has a weaker transmitter', with no evidence
- B** The one written by your best friend
- C** The one that is the longest
- D** The one that uses evidence from the stations, such as the direction of movement or a measurement

When you write a question for the Driving Question Board, which question best fits the instruction 'start with HOW or WHY, about the phenomenon, and something you cannot yet answer'?

- A** Where is the Kabete underpass on the map?
- B** Is Classic 105 a good radio station?
- C** Which Sacco owns the matatu?
- D** Why does the FM signal cut out in the tunnel while the AM signal does not?

You are asked to draw an 'Initial Wave Model' of a wave travelling from a matatu horn to your ear and entering a tunnel. Which statement about this model is correct?

- A** It only needs to show your current thinking, with labels and arrows, because you will revise it in later lessons.
- B** The teacher will mark it wrong if the wave shape is not right.
- C** It should copy the teacher's example exactly.
- D** It must be completely correct before you hand it in.

At the end of this first lesson, your class has noticed patterns and posted questions about the matatu horn and the Kabete tunnel. What is the status of the driving question 'Why does the horn change pitch, and why does FM fade in the tunnel?'

- A** It is answered because the PhET simulation showed the pitch change.
- B** It is fully answered because the class voted on the best explanation.
- C** It is still open: the class has observations and first ideas, and will gather more evidence in later lessons.
- D** It is answered because the diagram shows the FM wavelength and the tunnel width.

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).